PHD DEFENSE · MICROBIAL ECOLOGY · JUNE 2026

Why some guts bounce back from antibiotics and others don't

A four-year study of microbiome recovery and the keystone species that drive it.

The question: a single antibiotic course can reshape the gut for years — but not in everyone.

A week of broad-spectrum antibiotics wipes out most gut bacteria in days. In most people the community rebuilds within weeks. In a minority it never fully returns. My thesis asks what separates the two — and whether we can predict, before treatment, which gut is fragile.

Why it matters: the guts that don't recover carry lasting health costs.

Incomplete recovery is linked to recurrent infection, metabolic dysregulation, and weakened immune response. If a clinician could flag a fragile microbiome before prescribing, antibiotic stewardship could be personalised rather than population-wide.

- Recurrent *C. difficile* infection tracks with poor recovery.
- No pre-treatment test currently exists.

Methods: a 200-person longitudinal cohort, sampled before, during, and for a year after.

We followed 200 adults prescribed a standard amoxicillin-clavulanate course. Stool was collected at baseline, day 7, day 30, and months 3, 6, and 12. Shotgun metagenomics gave species-resolution community profiles at every timepoint.

- 1,200 samples, deep-sequenced at species resolution.
- Recovery defined as Bray–Curtis return to within 0.2 of baseline.

A keystone framing: not all species matter equally to recovery.

I model the gut as an ecological network and ask which species, if absent at baseline, predict collapse. Borrowing from ecology, I treat recovery as a resilience property of the network — driven by a few keystone taxa, not by raw diversity.

- Resilience is a network property, not a headcount.
- The hypothesis: keystones, not diversity, drive recovery.

PRIMARY RESULT · 12-MONTH RECOVERY OUTCOMES

Recovery splits the cohort cleanly, and one genus explains most of it.

n = 200, full recovery defined at the 12-month timepoint.

68%
FULLY RECOVERED

32%
INCOMPLETE AT 12 MO

3.4×
RECOVERY ODDS WITH KEYSTONE PRESENT

p<0.001
KEYSTONE ASSOCIATION

The keystone is *Faecalibacterium prausnitzii* — present at baseline, recovery follows.

Participants who carried abundant *F. prausnitzii* before treatment recovered 3.4 times more often. Its baseline abundance predicted recovery better than total diversity, age, or antibiotic dose. The fragile guts were missing this one anaerobe going in.

- Baseline *F. prausnitzii* outperformed diversity as a predictor.
- The effect held after adjusting for age, BMI, and dose.

Validation: the predictor holds in an independent 80-person replication cohort.

A separate cohort, sampled two years later with a different sequencing platform, reproduced the association: baseline keystone abundance predicted recovery with 0.79 AUC. The signal is not an artefact of one batch or one machine.

- External AUC 0.79, close to the discovery cohort's 0.82.
- Different platform, different year, same effect.

Discussion: why a single anaerobe holds the network together.

F. prausnitzii is a major butyrate producer; butyrate feeds the gut lining and supports dozens of dependent species. Lose the keystone and the cross-feeding web that lets others rebuild collapses with it. Diversity is the symptom; the keystone is the cause.

- Butyrate is the currency the recovery network runs on.
- Cross-feeding makes the keystone load-bearing.

FUTURE WORK · 2026–2028

Where this goes next, by research track.

TRACK	NOW (2026)	NEXT (2027)	LATER (2028)
Prediction	✓ Keystone biomarker	– Clinical assay prototype	○ Point-of-care test
Intervention	✓ In-vitro restoration	– Pre-treatment supplementation trial	○ Randomised clinical trial
Mechanism	✓ Butyrate cross-feeding map	– Strain-level resolution	○ Causal gnotobiotic models

✓ Shipped – In flight ○ Planned

Limitations I want the committee to weigh.

The cohort is observational — I show association, not proof that adding the keystone causes recovery. The gnotobiotic models in track three are designed to close that gap. And recovery was defined by composition, not function; two communities can look different yet work the same.

- Causation awaits the gnotobiotic experiments.
- Compositional recovery may understate functional recovery.

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A fragile gut is one missing its keystone — and we can see it before the first dose.